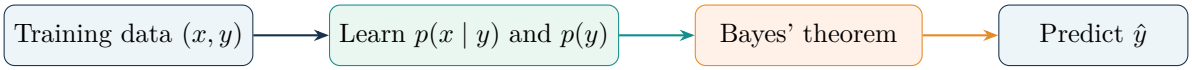
Generative Learning Algorithms

Complete Mathematical Derivations, Diagrams & Exam Notes

Study Notes



Abstract

These notes develop the main ideas of generative classification from first principles: Bayes' theorem, multivariate Gaussian distributions, Gaussian Discriminant Analysis (GDA), the link between GDA and logistic regression, Naive Bayes, Bernoulli and Multinomial text models, and Laplace smoothing. Derivations are shown step by step, with diagrams, worked examples, comparison tables, and exam-focused summaries.

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1 Generative versus discriminative learning

Let $x \in \mathbb{R}^d$ be a feature vector and $y \in \{0, 1\}$ a class label. For example, x might contain an animal's height and weight, while $y = 0$ means dog and $y = 1$ means elephant.

1.1 Two modelling approaches

- **Discriminative learning** directly models the conditional probability $p(y \mid x)$ or learns a decision boundary. Examples include logistic regression and the perceptron.
- **Generative learning** models the class prior $p(y)$ and class-conditional distribution $p(x \mid y)$. It then applies Bayes' theorem to compute $p(y \mid x)$. Examples include GDA and Naive Bayes.

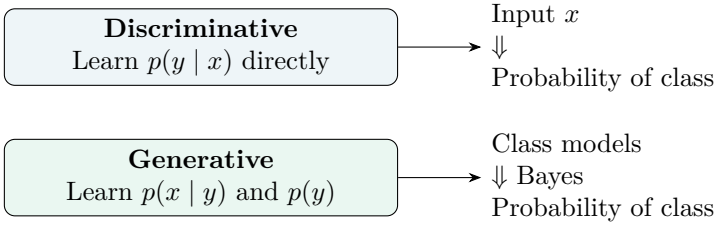


Figure 1: Discriminative and generative routes to classification.

1.2 Bayes' theorem and the classification rule

Bayes' theorem is

$$\boxed{p(y \mid x) = \frac{p(x \mid y)p(y)}{p(x)}}.$$

For binary classes, the law of total probability gives

$$p(x) = p(x \mid y = 0)p(y = 0) + p(x \mid y = 1)p(y = 1).$$

Therefore,

$$p(y = 1 \mid x) = \frac{p(x \mid y = 1)p(y = 1)}{p(x \mid y = 0)p(y = 0) + p(x \mid y = 1)p(y = 1)}.$$

The denominator is the same for every candidate class. Thus,

$$\boxed{\hat{y} = \arg\max_y p(y \mid x) = \arg\max_y p(x \mid y)p(y).}$$

The likelihood says how compatible the observed features are with a class; the prior says how common the class was before seeing the features.

1.3 Numerical example

Suppose

$$p(x \mid y = 0) = 0.2, \quad p(y = 0) = 0.7, \qquad p(x \mid y = 1) = 0.6, \quad p(y = 1) = 0.3.$$

The unnormalized scores are $S_0 = 0.2(0.7) = 0.14$ and $S_1 = 0.6(0.3) = 0.18$. Hence $p(x) = 0.32$ and

$$p(y = 1 \mid x) = \frac{0.18}{0.32} = 0.5625, \qquad p(y = 0 \mid x) = 0.4375.$$

Predict class 1, even though class 0 has the larger prior.

2 The multivariate Gaussian distribution

2.1 Definition and density

A scalar normal variable is $X \sim \mathcal{N}(\mu, \sigma^2)$. In d dimensions,

$$X \sim \mathcal{N}(\mu, \Sigma), \qquad \mu \in \mathbb{R}^d, \quad \Sigma \in \mathbb{R}^{d \times d}.$$

The multivariate Gaussian density is

$$\boxed{p(x; \mu, \Sigma) = \frac{1}{(2\pi)^{d/2}|\Sigma|^{1/2}} \exp\left[-\frac{1}{2}(x-\mu)^T \Sigma^{-1}(x-\mu)\right].}$$

Here $|\Sigma|$ is the determinant. For the usual nonsingular density, Σ is symmetric positive definite.

- μ is the mean vector and marks the centre of the distribution.
- Σ is the covariance matrix; its diagonal entries are variances and off-diagonal entries are covariances.
- $(x-\mu)^T \Sigma^{-1}(x-\mu)$ is the squared Mahalanobis distance.
- $(2\pi)^{d/2}|\Sigma|^{1/2}$ is the normalization factor that makes the density integrate to one.

2.2 Deriving the density by change of variables

Start with a standard normal vector

$$Z \sim \mathcal{N}(0, I_d), \qquad p_Z(z) = \frac{1}{(2\pi)^{d/2}} \exp\left(-\frac{1}{2} z^T z\right).$$

Transform it using $X = \mu + AZ$, where A is invertible. Then

$$z = A^{-1}(x - \mu).$$

The change-of-variables rule gives

$$p_X(x) = p_Z(A^{-1}(x-\mu)) \left|\det(A^{-1})\right|.$$

Substitution yields

$$p_X(x) = \frac{1}{(2\pi)^{d/2}|\det A|} \exp\left[-\frac{1}{2}(x-\mu)^T A^{-T} A^{-1}(x-\mu)\right].$$

The covariance of X is

$$\mathrm{Cov}(X) = A\,\mathrm{Cov}(Z)A^T = AA^T.$$

Set $\Sigma = AA^T$. Then $\Sigma^{-1} = A^{-T}A^{-1}$ and $|\Sigma| = |\det A|^2$, so $|\det A| = |\Sigma|^{1/2}$. The Gaussian density follows.

2.3 Mean derivation

Since $X = \mu + AZ$ and $\mathbb{E}[Z] = 0$,

$$\mathbb{E}[X] = \mathbb{E}[\mu + AZ] = \mu + A\mathbb{E}[Z] = \boxed{\mu}.$$

2.4 Covariance derivation

By definition,

$$\mathrm{Cov}(X) = \mathbb{E}[(X - \mathbb{E}[X])(X - \mathbb{E}[X])^T].$$

Since $X - \mathbb{E}[X] = AZ$,

$$\begin{aligned}\mathrm{Cov}(X) &= \mathbb{E}[(AZ)(AZ)^T] \\ &= A\mathbb{E}[ZZ^T]A^T \\ &= AIA^T = \boxed{\Sigma}.\end{aligned}$$

Another useful identity is

$$\boxed{\mathrm{Cov}(X) = \mathbb{E}[XX^T] - \mathbb{E}[X]\mathbb{E}[X]^T.}$$

To verify it, expand $\mathbb{E}[(X - \mu)(X - \mu)^T]$ and use $\mathbb{E}[X] = \mu$.

2.5 Covariance matrix and geometry

For two features,

$$\Sigma = \begin{bmatrix} \sigma_1^2 & \sigma_{12} \\ \sigma_{12} & \sigma_2^2 \end{bmatrix}.$$

The diagonal contains variances; the off-diagonal entries contain covariance. Positive covariance indicates a tendency for the two variables to move together, while negative covariance indicates an opposite tendency. Zero covariance does not generally imply independence, but it does for jointly Gaussian variables.

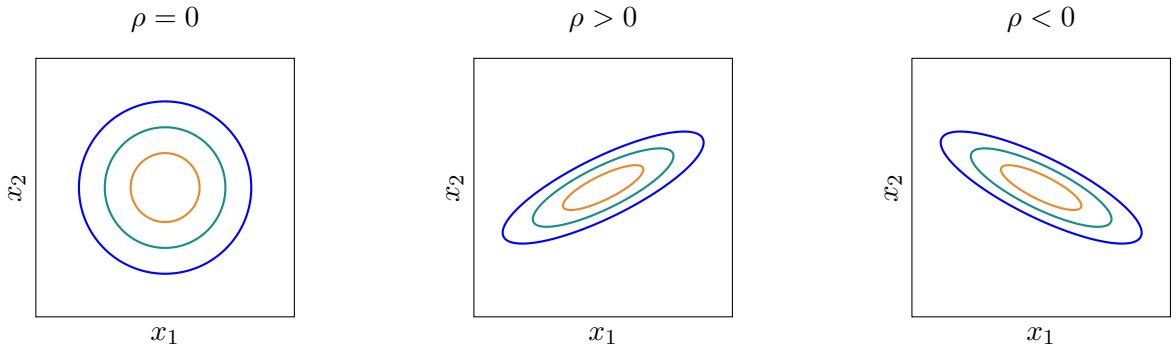


Figure 2: Schematic Gaussian contours: circular for no correlation, elongated along the positive diagonal for positive correlation, and along the negative diagonal for negative correlation.

For $\Sigma = \begin{bmatrix} 1 & \rho \\ \rho & 1 \end{bmatrix}$, the eigenvalues are $1 + \rho$ and $1 - \rho$, with eigenvectors along $(1, 1)^T$ and $(1, -1)^T$. As $|\rho|$ approaches one, one direction becomes much narrower and the contour becomes elongated.

2.6 Effect of mean and covariance

Changing μ translates the distribution without changing its shape. Increasing covariance spreads probability mass over a larger region and lowers the peak; decreasing covariance concentrates the mass and raises the peak. In two dimensions, equal-density contours are ellipses.

3 Gaussian Discriminant Analysis (GDA)

3.1 Model assumptions

GDA models continuous features with class-conditional Gaussians:

$$y \sim \text{Bernoulli}(\phi), \qquad x \mid y = 0 \sim \mathcal{N}(\mu_0, \Sigma), \qquad x \mid y = 1 \sim \mathcal{N}(\mu_1, \Sigma).$$

The parameters are $\phi, \mu_0, \mu_1, \Sigma$. The key assumption is a *shared* covariance matrix, while the class means may differ.

The class prior is

$$p(y) = \phi^y (1-\phi)^{1-y}.$$

The class-conditional density for class k is

$$p(x \mid y = k) = \frac{1}{(2\pi)^{d/2}|\Sigma|^{1/2}} \exp\left[-\frac{1}{2}(x-\mu_k)^T \Sigma^{-1}(x-\mu_k)\right].$$

3.2 Likelihood and log-likelihood

For i.i.d. data $\{(x^{(i)}, y^{(i)})\}_{i=1}^n$,

$$L(\phi, \mu_0, \mu_1, \Sigma) = \prod_{i=1}^n p(x^{(i)} \mid y^{(i)}) p(y^{(i)}).$$

Taking logarithms,

$$\begin{aligned} \ell = & -\frac{nd}{2}\log(2\pi) - \frac{n}{2}\log|\Sigma| \\ & -\frac{1}{2}\sum_{i=1}^n (x^{(i)} - \mu_{y^{(i)}})^T \Sigma^{-1}(x^{(i)} - \mu_{y^{(i)}}) \\ & +\sum_{i=1}^n \left[y^{(i)}\log\phi + (1-y^{(i)})\log(1-\phi)\right]. \end{aligned}$$

3.3 MLE for the class prior ϕ

Keep only terms involving ϕ:

$$\ell(\phi) = \sum_{i=1}^n \left[y^{(i)}\log\phi + (1-y^{(i)})\log(1-\phi)\right].$$

Differentiate and set to zero:

$$\frac{\partial \ell}{\partial \phi} = \frac{\sum_i y^{(i)}}{\phi} - \frac{n - \sum_i y^{(i)}}{1-\phi} = 0.$$

Cross-multiplying gives

$$(1-\phi)\sum_i y^{(i)} = \phi\left(n - \sum_i y^{(i)}\right).$$

After simplification, $\sum_i y^{(i)} = n\phi$, so

$$\boxed{\hat{\phi} = \frac{1}{n}\sum_{i=1}^n y^{(i)} = \frac{\#\{i : y^{(i)} = 1\}}{n}.}$$

3.4 MLE for class means

For class k, let $I_k = \{i : y^{(i)} = k\}$ and $n_k = |I_k|$. Terms involving μ_k are

$$\ell(\mu_k) = -\frac{1}{2}\sum_{i \in I_k}(x^{(i)} - \mu_k)^T \Sigma^{-1}(x^{(i)} - \mu_k) + C.$$

Using the matrix derivative

$$\frac{\partial}{\partial \mu}(x-\mu)^T A(x-\mu) = -2A(x-\mu)$$

for symmetric A, we get

$$\frac{\partial \ell}{\partial \mu_k} = \sum_{i \in I_k} \Sigma^{-1}(x^{(i)} - \mu_k) = 0.$$

Multiply by Σ:

$$\sum_{i \in I_k} x^{(i)} - n_k \mu_k = 0.$$

Therefore,

$$\boxed{\hat{\mu}_k = \frac{1}{n_k}\sum_{i:y^{(i)}=k} x^{(i)}.}$$

Each class mean is the average feature vector of examples in that class.

3.5 MLE for the shared covariance

Let $r_i = x^{(i)} - \mu_{y^{(i)}}$ and $S = \sum_i r_i r_i^T$. The covariance-dependent log-likelihood is

$$\ell(\Sigma) = -\frac{n}{2}\log|\Sigma| - \frac{1}{2}\operatorname{tr}(\Sigma^{-1}S) + C.$$

Use

$$\frac{\partial \log|\Sigma|}{\partial \Sigma} = \Sigma^{-1}, \qquad \frac{\partial \operatorname{tr}(\Sigma^{-1}S)}{\partial \Sigma} = -\Sigma^{-1}S\Sigma^{-1}.$$

Then

$$\frac{\partial \ell}{\partial \Sigma} = -\frac{n}{2}\Sigma^{-1} + \frac{1}{2}\Sigma^{-1}S\Sigma^{-1} = 0.$$

Multiplying on both sides by Σ gives $S = n\Sigma$. Thus,

$$\boxed{\hat{\Sigma} = \frac{1}{n}\sum_{i=1}^{n}(x^{(i)} - \hat{\mu}_{y^{(i)}})(x^{(i)} - \hat{\mu}_{y^{(i)}})^T.}$$

This is the maximum-likelihood estimate and uses denominator n.

GDA MLE summary: $\hat{\phi} = \frac{1}{n}\sum_i \mathbf{1}\{y^{(i)} = 1\}$, $\hat{\mu}_k = \frac{1}{n_k}\sum_{i:y^{(i)}=k} x^{(i)}$, $\hat{\Sigma} = \frac{1}{n}\sum_i (x^{(i)} - \hat{\mu}_{y^{(i)}})(x^{(i)} - \hat{\mu}_{y^{(i)}})^T$.

3.6 Why GDA has a linear decision boundary

Predict class 1 when

$$p(x \mid y=1)\phi > p(x \mid y=0)(1-\phi).$$

Taking logs,

$$\log p(x \mid y=1) - \log p(x \mid y=0) + \log\frac{\phi}{1-\phi} > 0.$$

The common Gaussian normalization terms cancel. Expanding the quadratic form,

$$(x-\mu_k)^T\Sigma^{-1}(x-\mu_k) = x^T\Sigma^{-1}x - 2\mu_k^T\Sigma^{-1}x + \mu_k^T\Sigma^{-1}\mu_k.$$

The term $x^T\Sigma^{-1}x$ cancels between classes. Rearranging yields

$$\boxed{\theta^T x + b > 0}$$

with

$$\boxed{\theta = \Sigma^{-1}(\mu_1 - \mu_0)}$$

and

$$\boxed{b = -\frac{1}{2}\mu_1^T\Sigma^{-1}\mu_1 + \frac{1}{2}\mu_0^T\Sigma^{-1}\mu_0 + \log\frac{\phi}{1-\phi}.}$$

Hence the decision boundary is the hyperplane $\theta^T x + b = 0$. If class-specific covariance matrices differ, the quadratic terms do not cancel in general; this leads to quadratic discriminant analysis (QDA).

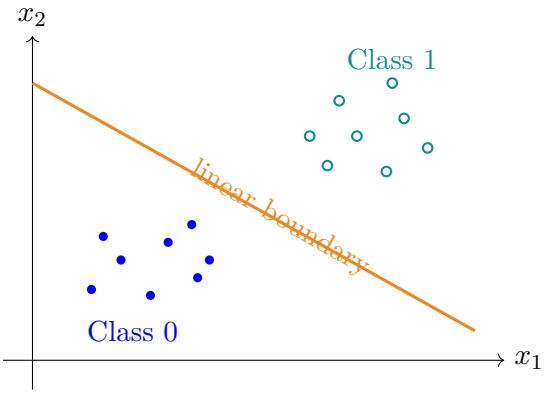


Figure 3: Schematic GDA classification: Gaussian class clusters and a linear decision boundary.

4 GDA and logistic regression

4.1 Deriving the logistic posterior

Start from the posterior odds:

$$\frac{p(y=1 \mid x)}{p(y=0 \mid x)} = \frac{p(x \mid y=1)\phi}{p(x \mid y=0)(1-\phi)}.$$

Taking logarithms and using the shared-covariance Gaussian model,

$$\log\frac{p(y=1 \mid x)}{1-p(y=1 \mid x)} = \theta^T x + b.$$

Let $p = p(y=1 \mid x)$. Then

$$\log\frac{p}{1-p} = \theta^T x + b \quad \Longrightarrow \quad \frac{p}{1-p} = e^{\theta^T x + b}.$$

Solving for p,

$$p(1 + e^{\theta^T x + b}) = e^{\theta^T x + b}$$

and therefore

$$\boxed{p(y=1 \mid x) = \frac{1}{1+e^{-(\theta^T x + b)}} = \sigma(\theta^T x + b).}$$

This is the logistic form. GDA derives it from a generative Gaussian assumption; logistic regression assumes the conditional probability has this form and estimates its parameters directly.

4.2 Comparison

Property	GDA	Logistic regression
Type	Generative	Discriminative
Direct model	$p(x \mid y)$ and $p(y)$	$p(y \mid x)$
Assumption	Gaussian class-conditional features with shared covariance	Logistic conditional probability
Boundary	Linear under shared covariance	Linear in input features
When assumptions hold	Can be statistically efficient	May need more data for comparable estimation accuracy
When assumptions fail	Sensitive to distribution mismatch	Often more robust to incorrect assumptions about $p(x \mid y)$

Neither is always better. The choice depends on whether the generative assumptions are plausible, sample size, feature structure, and regularization.

5 Naive Bayes

5.1 Conditional independence

For a feature vector $x = (x_1, \ldots, x_d)$, the probability chain rule gives

$$p(x_1, \ldots, x_d \mid y) = p(x_1 \mid y)p(x_2 \mid x_1, y) \cdots p(x_d \mid x_1, \ldots, x_{d-1}, y).$$

Naive Bayes assumes that the features are conditionally independent given y, so

$$p(x_j \mid x_1, \ldots, x_{j-1}, y) = p(x_j \mid y).$$

Therefore,

$$\boxed{p(x \mid y) = \prod_{j=1}^{d} p(x_j \mid y).}$$

This does *not* assert unconditional independence. Features can be correlated overall because the class label affects their distributions.

5.2 Prediction rule and log scores

Bayes' theorem gives

$$p(y \mid x) = \frac{p(y) \prod_{j=1}^{d} p(x_j \mid y)}{p(x)}.$$

Since $p(x)$ is constant across candidate labels,

$$\boxed{\hat{y} = \arg\max_y p(y) \prod_{j=1}^{d} p(x_j \mid y).}$$

Products of many small numbers can underflow numerically. Taking logarithms preserves the maximizing class:

$$\boxed{\hat{y} = \arg\max_y \left[\log p(y) + \sum_{j=1}^{d} \log p(x_j \mid y) \right].}$$

6 Bernoulli Naive Bayes

6.1 Binary features and probability mass function

In text classification, define $x_j = 1$ if vocabulary word j occurs in the email, and $x_j = 0$ otherwise. Let

$$\phi_{j|k} = p(x_j = 1 \mid y = k).$$

A Bernoulli variable has probability $\phi_{j|k}$ for 1 and $1 - \phi_{j|k}$ for 0. Both cases are captured by

$$\boxed{p(x_j \mid y = k) = \phi_{j|k}^{x_j}(1 - \phi_{j|k})^{1-x_j}.}$$

Under the Naive Bayes assumption,

$$\boxed{p(x \mid y = k) = \prod_{j=1}^{|V|} \phi_{j|k}^{x_j}(1 - \phi_{j|k})^{1-x_j}.}$$

This model explicitly accounts for absent words.

6.2 MLE derivation for a Bernoulli feature

Fix word j and class k. Let N_k be the number of class-k documents and c_{jk} the number containing word j. The likelihood is

$$L(\phi_{j|k}) = \phi_{j|k}^{c_{jk}}(1 - \phi_{j|k})^{N_k - c_{jk}}.$$

The log-likelihood is

$$\ell = c_{jk} \log \phi_{j|k} + (N_k - c_{jk}) \log(1 - \phi_{j|k}).$$

Differentiate and set to zero:

$$\frac{c_{jk}}{\phi_{j|k}} - \frac{N_k - c_{jk}}{1 - \phi_{j|k}} = 0.$$

Cross-multiplying gives $c_{jk} = N_k \phi_{j|k}$, so

$$\boxed{\hat{\phi}_{j|k} = \frac{c_{jk}}{N_k}.}$$

The class-prior estimate is

$$\boxed{\hat{\phi}_y = \frac{1}{n} \sum_{i=1}^{n} \mathbf{1}\{y^{(i)} = 1\}.}$$

6.3 Bernoulli prediction

For each class k, define the log score

$$S_k = \log p(y = k) + \sum_{j=1}^{|V|} \Big[x_j \log \phi_{j|k} + (1 - x_j) \log(1 - \phi_{j|k}) \Big].$$

Then

$$\boxed{\hat{y} = \arg\max_{k \in \{0,1\}} S_k.}$$

7 Laplace smoothing

7.1 The zero-frequency problem

If a word has never appeared in training examples of a class, its MLE is zero. Since Naive Bayes multiplies feature probabilities, a single zero factor makes the entire class likelihood zero. If every class likelihood becomes zero for a test example, Bayes' formula can become undefined.

7.2 Derivation of the smoothed estimate

Suppose a categorical variable has K possible outcomes. Let c_j be the count for outcome j, and $N = \sum_j c_j$. The unsmoothed MLE is $\hat{p}_j = c_j/N$. Add one pseudo-count to every outcome:

$$\boxed{\hat{p}_j = \frac{c_j + 1}{N + K}.}$$

The probabilities still sum to one:

$$\sum_{j=1}^{K} \hat{p}_j = \frac{\sum_j c_j + K}{N + K} = 1.$$

This is also the posterior mean under a symmetric Dirichlet$(1, \ldots, 1)$ prior.

7.3 Bernoulli smoothing

Each Bernoulli feature has two outcomes (present and absent), so

$$\boxed{\hat{\phi}_{j|k} = \frac{c_{jk} + 1}{N_k + 2}.}$$

The denominator adds 2 because there are two outcomes per feature.

7.4 Multinomial smoothing

A categorical word-token variable has $|V|$ possible outcomes, so

$$\boxed{\hat{\phi}_{w|y} = \frac{c_{wy} + 1}{N_y + |V|}.}$$

Here c_{wy} is the number of occurrences of word w in class y, and N_y is the total number of word tokens in that class.

8 Bernoulli versus Multinomial event models

8.1 Bernoulli event model

The feature vector records whether each vocabulary word appears:

$$x_j \in \{0, 1\}, \qquad j = 1, \ldots, |V|.$$

Repeated occurrences do not change a feature from 1. The document likelihood is

$$p(x \mid y) = \prod_{j=1}^{|V|} \phi_{j|y}^{x_j} (1 - \phi_{j|y})^{1 - x_j}.$$

Absence of a word is informative and contributes a factor.

8.2 Multinomial event model

A document is a sequence of word tokens. If x_j is the identity of token j, then

$$p(x \mid y) = \prod_{j=1}^{d} p(x_j \mid y).$$

Let $\phi_{w|y} = p(\text{token} = w \mid y)$, with

$$\phi_{w|y} \geq 0, \qquad \sum_{w \in V} \phi_{w|y} = 1.$$

The class likelihood is proportional to

$$L(\boldsymbol{\phi}_y) = \prod_{w \in V} \phi_{w|y}^{c_{wy}}.$$

8.3 Deriving the Multinomial MLE

Take logs:

$$\ell(\boldsymbol{\phi}_y) = \sum_{w \in V} c_{wy} \log \phi_{w|y}.$$

Maximize subject to $\sum_w \phi_{w|y} = 1$ using a Lagrange multiplier:

$$\mathcal{J} = \sum_w c_{wy} \log \phi_{w|y} + \lambda \left(\sum_w \phi_{w|y} - 1 \right).$$

Differentiate:

$$\frac{c_{wy}}{\phi_{w|y}} + \lambda = 0 \quad \Longrightarrow \quad \phi_{w|y} = -\frac{c_{wy}}{\lambda}.$$

Summing over w and using $\sum_w \phi_{w|y} = 1$ gives $\lambda = -N_y$, where $N_y = \sum_w c_{wy}$. Thus,

$$\boxed{\hat{\phi}_{w|y} = \frac{c_{wy}}{N_y}.}$$

With Laplace smoothing,

$$\boxed{\hat{\phi}_{w|y} = \frac{c_{wy} + 1}{N_y + |V|}.}$$

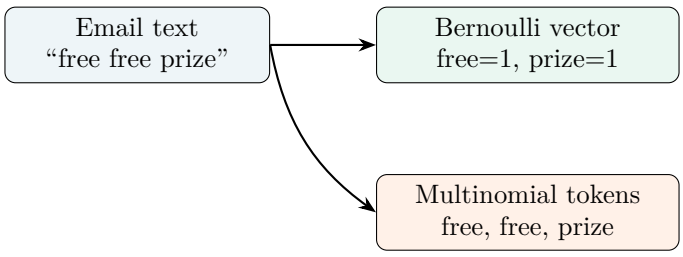


Figure 4: Bernoulli records presence/absence; Multinomial retains repeated word tokens.

9 Worked numerical example: Bernoulli Naive Bayes

Let the vocabulary be $\{\text{free}, \text{win}, \text{meeting}\}$. Suppose

$$p(y = 1) = 0.4, \qquad p(y = 0) = 0.6,$$

and the word-presence probabilities are

	free	win	meeting
$y = 1$	0.8	0.6	0.1
$y = 0$	0.1	0.05	0.7

A new email contains "free" and "win", but not "meeting", so $x = (1, 1, 0)$.

Spam score:

$$S_1 = 0.4(0.8)(0.6)(1 - 0.1) = \boxed{0.1728}.$$

Non-spam score:

$$S_0 = 0.6(0.1)(0.05)(1 - 0.7) = \boxed{0.0009}.$$

Normalize:

$$p(y = 1 \mid x) = \frac{0.1728}{0.1728 + 0.0009} \approx \boxed{0.9948}.$$

Thus, predict spam. Notice that the absent word "meeting" contributes a factor in both class scores.

10 Master comparison table

Property	GDA	Bernoulli NB	Multinomial NB
Typical features	Continuous	Binary presence/absence	Word tokens/counts
Core assumption	Gaussian class distributions; shared covariance	Conditional independence of binary features	Conditional independence of word tokens
Parameters	Means, covariance, prior	Presence probabilities, prior	Token probabilities, prior
Do absent words matter?	Not in this representation	Yes	No explicit absence factor
Smoothing issue	Singular covariance can be a problem	Zero-frequency estimates	Zero-frequency estimates
Common remedy	Covariance regularization	Laplace smoothing	Laplace smoothing

11 Common exam mistakes

1. GDA's shared covariance does not mean shared class means.
2. The covariance MLE divides by n; an unbiased sample estimator uses a different denominator.
3. Shared covariance makes the GDA boundary linear. Different class covariances generally yield a quadratic boundary.
4. Naive Bayes assumes conditional, not necessarily unconditional, independence.
5. Bernoulli Naive Bayes includes absent words; Multinomial Naive Bayes models token identity/frequency.
6. Bernoulli smoothing uses denominator $N_k + 2$; Multinomial smoothing uses $N_y + |V|$.
7. Log probabilities avoid numerical underflow and preserve the maximizing class.
8. A covariance matrix must be positive semidefinite; the ordinary Gaussian density formula with Σ^{-1} requires a nonsingular covariance.

12 One-page formula sheet

Bayes: $p(y \mid x) = \dfrac{p(x \mid y)p(y)}{p(x)}, \quad \hat{y} = \arg\max_y p(x \mid y)p(y)$

Gaussian: $p(x; \mu, \Sigma) = \dfrac{\exp[-\frac{1}{2}(x-\mu)^T\Sigma^{-1}(x-\mu)]}{(2\pi)^{d/2}|\Sigma|^{1/2}}$

$\mathbb{E}[X] = \mu, \qquad \mathrm{Cov}(X) = \Sigma$

GDA MLE: $\hat{\phi} = \dfrac{1}{n}\sum_i \mathbf{1}\{y^{(i)} = 1\}, \quad \hat{\mu}_k = \dfrac{1}{n_k}\sum_{i:y^{(i)}=k} x^{(i)}$

$\hat{\Sigma} = \dfrac{1}{n}\sum_i (x^{(i)} - \hat{\mu}_{y^{(i)}})(x^{(i)} - \hat{\mu}_{y^{(i)}})^T$

GDA boundary: $\theta = \Sigma^{-1}(\mu_1 - \mu_0), \quad b = -\dfrac{1}{2}\mu_1^T\Sigma^{-1}\mu_1 + \dfrac{1}{2}\mu_0^T\Sigma^{-1}\mu_0 + \log\dfrac{\phi}{1-\phi}$

$p(y = 1 \mid x) = \sigma(\theta^T x + b), \quad \sigma(z) = \dfrac{1}{1+e^{-z}}$

Naive Bayes: $p(x \mid y) = \prod_j p(x_j \mid y), \quad \hat{y} = \arg\max_y \left[\log p(y) + \sum_j \log p(x_j \mid y)\right]$

Bernoulli NB: $p(x_j \mid y = k) = \phi_{j|k}^{x_j}(1-\phi_{j|k})^{1-x_j}, \quad \hat{\phi}_{j|k} = \dfrac{c_{jk}}{N_k}$

Laplace: $\hat{\phi}_{j|k} = \dfrac{c_{jk}+1}{N_k+2}$

Multinomial NB: $\hat{\phi}_{w|y} = \dfrac{c_{wy}}{N_y}, \quad$ Laplace: $\hat{\phi}_{w|y} = \dfrac{c_{wy}+1}{N_y+|V|}$

Categorical smoothing: $\hat{p}_j = \dfrac{c_j+1}{N+K}.$

Final study strategy

1. Understand Bayes' theorem and why the evidence cancels in an arg max decision.
2. Learn the Gaussian density, the role of μ and Σ, and the geometry of covariance.
3. Re-derive the three GDA MLEs: prior, means, and shared covariance.
4. Derive the linear GDA boundary and then the sigmoid posterior.
5. Understand conditional independence before memorizing the Naive Bayes product.
6. Practise Bernoulli MLE, zero-frequency failure, and Laplace smoothing.
7. Finish by deriving Multinomial MLE with a Lagrange multiplier.

Learn the assumptions; derive the formulas.